

Working with nodes in JavaScript

As you know, tags can contain other tags or, in JavaScript terms, DOM elements can contain other DOM elements. In addition, however, they may contain comments and plain text. Comments, texts and DOM elements are united by one name - *nodes*.

You already know the properties `firstElementChild`, `lastElementChild`, `nextElementSibling`, `previousElementSibling`. These properties only work with DOM elements, ignoring other nodes. As a rule, this is exactly what we need.

However, there are also properties `firstChild`, `lastChild`, `nextSibling`, `previousSibling`. These properties work in a similar way, however, considering all nodes. Let's see the difference between these properties with an example. Suppose we have a div containing three nodes:

```
1 <div id="elem"><!--com-->text<span>tag</span></div>
```

Get a reference to this div into a variable:

```
1 let elem = document.querySelector('#elem');
```

Now let's see what is contained in the properties `firstChild` and `firstElementChild`:

```
1 console.log(elem.firstChild); // comment
2 console.log(elem.firstElementChild); // span tag
```

№ 1

⊗jsPmNdInr

Write code that shows the difference between `lastChild` and `lastElementChild`.

№ 2

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Write code that shows the difference between `nextSibling` and `nextElementSibling`.

№ 3

⊗jsPmNdInr

Write code that shows the difference between `previousSibling` and `previousElementSibling`.

Loop iteration of nodes in JavaScript

As you should already know, the property `children` contains all DOM elements that are direct children of this element. There is a similar property `childNodes` which contains all child nodes of an element.

Let's, for example, use this property to display various nodes from our element:

```
1 console.log(elem.childNodes[0]);
2 console.log(elem.childNodes[1]);
3 console.log(elem.childNodes[2]);
```

And now let's iterate over the nodes of the element in a loop:

```
1 for (let node of elem.childNodes) {
2   console.log(node);
3 }
```

Given a div:

```
1 | <div id="elem">txt<b>tag</b><!--com-->txt<b>tag</b><!--com--></div>
```

Loop through all the nodes of this div and print them to the console.

Node name in JavaScript

All nodes have the property `nodeName`. For a comment, it contains the value 'comment', for a text node, the value 'text', and for an element node, the tag name in uppercase. It turns out that `nodeName` for elements is similar to the property `tagName`, but works for other types of nodes as well.

Given a div:

```
1 | <div id="elem">txt<b>tag</b><!--com-->txt<b>tag</b><!--com--></div>
```

Loop through all the nodes of this div and print their names to the console.

Node type in JavaScript

All nodes have the property `nodeType`. Its value is numeric: 1 for elements, 3 for text nodes, 8 for comments. The remaining values are currently either little used or outdated.

Given a div:

```
1 | <div id="elem">txt<b>tag</b><!--com-->txt<b>tag</b><!--com--></div>
```

Loop through all the nodes of this div and output only the element nodes and text nodes to the console.

Node text in JavaScript

Let's now figure out how to read or change the text of the nodes. For all nodes, the `textContent` property works. For text nodes and comments, there are properties `nodeValue` and `data` (they are almost the same, the second one is shorter, it's better to use it). For elements there is a property `innerHTML` that reads the text along with the tags.

Given a div:

```
1 | <div id="elem">txt<b>tag</b><!--com-->txt<b>tag</b><!--com--></div>
```

Loop through all the nodes of this div and output the texts of all nodes to the console.

№2

⊗jsPmNdTx

Given a div:

```
1 | <div id="elem">txt<b>tag</b><!--com-->txt<b>tag</b><!--com--></div>
```

Loop through all nodes of this div and print to the console the texts of all comments and text nodes.

№3

⊗jsPmNdTx

Given a div:

```
1 | <div id="elem">txt<b>tag</b><!--com-->txt<b>tag</b><!--com--></div>
```

Loop through all the nodes of this div and output to the console the texts of the text nodes and elements.

№4

⊗jsPmNdTx

Given a div:

```
1 | <div id="elem">txt<b>tag</b><!--com-->txt<b>tag</b><!--com--></div>
```

Loop through all the nodes of this div and write down the type of each node to the its end.